Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 103
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

An ecommerce website uses an AWS Lambda function and an Amazon RDS for MySQL database for an order fulfillment service. The service needs to return order confirmation immediately.

During a marketing campaign that caused an increase in the number of orders, the website's operations team noticed errors for “too many connections” from Amazon RDS. However, the RDS DB cluster metrics are healthy. CPU and memory capacity are still available.

What should a developer do to resolve the errors?

A. Initialize the database connection outside the handler function. Increase the max_user_connections value on the parameter group of the DB cluster. Restart the DB cluster.

B. Initialize the database connection outside the handler function. Use RDS Proxy instead of connecting directly to the DB cluster.

C. Use Amazon Simple Queue Service (Amazon SQS) FIFO queues to queue the orders. Ingest the orders into the database. Set the Lambda function's concurrency to a value that equals the number of available database connections.

D. Use Amazon Simple Queue Service (Amazon SQS) FIFO queues to queue the orders. Ingest the orders into the database. Set the Lambda function's concurrency to a value that is less than the number of available database connections.

Show Suggested Answer

by MrTee at *April 22, 2023, 3:39 p.m.*

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Comments

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: B

Use an RDS Proxy instead of connecting directly to the DB cluster.

upvoted 13 times

hanJR Highly Voted 2 years, 11 months ago

B

https://aws.amazon.com/blogs/compute/using-amazon-rds-proxy-with-aws-lambda/

upvoted 7 times

sumanshu Most Recent 1 year, 3 months ago

Selected Answer: B

The "too many connections" error occurs because Lambda functions create separate database connections for each invocation.
RDS Proxy is designed to manage database connections efficiently by pooling and reusing connections, reducing the load on the database. Lambda functions can share connections from the proxy pool, avoiding the "too many connections" issue.

A) Eliminated - Increasing max_user_connections temporarily alleviates the issue but does not address the root cause

upvoted 3 times

Saudis 1 year, 6 months ago

Selected Answer: B

The key word is increase in the number of orders, so the proxy can help in this stuation

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: B

B is the correct answer.

upvoted 1 times

maurice2005 2 years ago

Selected Answer: C

Option B only improve the execution time of lambda and decrease the delay from request to database. It might even worsen the situation because database can get more concurrent connection. RDS Proxy also doesn't limit the number of connections, even if so it will generate errors for lambda.

The only way is to throttle the requests using SQS until a connection gets released.
Why everyone thinks ChatGPT has the ultimate answer????!!!!

upvoted 2 times

nbxyzd 1 year, 5 months ago

Read carefully: "the service needs to return order confirmation *immediately*."
This means that asynchronous solutions like SQS does NOT meet the requirement.

upvoted 2 times

SerialiDr 2 years, 2 months ago

Selected Answer: B

AWS RDS Proxy is designed to manage and pool database connections, which makes it ideal for environments with highly variable and potentially high-volume database access patterns, such as those driven by Lambda functions. It helps to reduce the number of direct connections to the database and can efficiently manage the connections from the pool.

upvoted 3 times

hmdev 2 years, 7 months ago

Selected Answer: B

We can use an RDS proxy to handle a lot of connections. We are choosing this option because the load on the RDS is normal. If the RDS was unable to handle loads, we would've checked other options like queues or transactions.

upvoted 2 times

eberhe900 2 years, 9 months ago

Selected Answer: B

https://repost.aws/questions/QULXSqEPGbQx6qiyBa1D1Udg/lambda-to-db-connectivity-best-practices

upvoted 1 times

loctong 2 years, 10 months ago

Selected Answer: B

Using an RDS Proxy can manage connections to the RDS instance, reducing the overhead of establishing new connections and thereby preventing the "too many connections" error.

upvoted 2 times